

Working Notes on Chiral Koszul Duality

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These notes are about what happens when you compute with a 1-form, what it teaches you when you fail, and why the failures contain more information than the successes.

Contents

1	A 1-form	2
2	The trouble with two poles	2
3	Three letters d	3
4	Fourier	4
5	Nilpotence is the logarithm of periodicity	4
6	Two things called E_1	6
7	The BV complex is the bar complex	7
8	Koszulness is a theorem, not a definition	8
9	What the bar complex counts	8
10	Five invariants, one number	9
11	The frame atom	10
12	Category $\mathcal{O}$	10
13	The frontier	11
14	What the 1-form knows	11
15	NF-1: the H^4 barrier	12
15.1	What is proved	12
15.2	Where we are stuck: the $R^{(1)}$ barrier	13
15.3	The PBW paradox and its resolution	14
15.4	What remains for NF-1	14

15.5	The 84	15
16	Frontier assessment, March 2026	15
16.1	What is proved (Stratum I)	15
16.2	Where we are stuck	16
16.3	Concrete insights from computation	16
16.4	Next moves	17
16.5	The shape of the problem	17

I A 1-form

$$\eta_{ij} = d \log(z_i - z_j). \quad (1)$$

A 1-form on the configuration space of two distinct points on a curve. It has a simple pole along $z_i = z_j$, with residue 1. Not residue k , not residue depending on anything. Just 1.

A double pole $dz/(z-w)^2$ has no coordinate-independent residue. A regular form extracts nothing. Only the logarithmic derivative threads the needle: a pole strong enough to detect, mild enough to be canonical.

For three points:

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0. \quad (2)$$

This is Arnold's relation. It holds because $(z_1 - z_2)(z_2 - z_3) + (z_2 - z_3)(z_3 - z_1) + (z_3 - z_1)(z_1 - z_2) = 0$, which holds because addition is commutative. The deepest identity in the subject follows from the fact that $a + b + c = 0$ implies $ab + bc + ca = 0$ when $a + b + c = 0$. It is the only relation.

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Now place elements $\phi_1, \dots, \phi_n$ of a chiral algebra $\mathcal{A}$ at n points on a smooth curve X . Dress them in logarithmic forms. Define a differential by extracting residues at collisions. Arnold's relation kills every triple residue. So $d^2 = 0$, and you have a cochain complex: the **bar complex** $\bar{B}_X(\mathcal{A})$.

Under a coordinate change $z \mapsto w(z)$, the propagator picks up a correction: $\eta_{ij} \mapsto \eta_{ij} + d \log \frac{w(z_i) - w(z_j)}{z_i - z_j}$. Near the diagonal this becomes $d \log w'(z_i)$, which is the standard cocycle for $\text{Diff}(X)$. The propagator transforms as a BRST ghost for diffeomorphisms.

2 The trouble with two poles

The Heisenberg OPE $\alpha(z)\alpha(w) = k/(z-w)^2$ has one pole. The bar differential extracts a scalar and stops. Everything works. Cohomology counts partitions. This should be a warning.

Kac-Moody has two poles:

$$J^a(z)J^b(w) = \frac{k\kappa^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w} + \text{reg.}$$

The simple pole is the Lie bracket. The double pole is the level. So the bar differential splits: $d = d_{\text{bracket}} + d_{\text{level}}$.

$d_{\text{bracket}}^2 \neq 0$ whenever the OPE has both poles. This holds across all 2048 sign conventions arising from eleven independent binary choices in the bar construction (grading, suspension, contraction, Koszul signs, etc.). No convention saves it.

What happens: iterated residues at a triple collision produce cross-terms between the bracket and the level. Jacobi cancels the bracket-bracket part. The bracket-level cross-terms are proportional to k and survive. The relation that actually holds is

$$d_{\text{bracket}}^2 = -\{d_{\text{bracket}}, d_{\text{level}}\}. \quad (3)$$

The pair $(d_{\text{bracket}}, d_{\text{level}})$ is not a bicomplex. The total d squares to zero by the Borchers identity, which is not the Jacobi identity with extra terms. It is the algebraic shadow of Stokes' theorem on the Fulton–MacPherson compactification $\overline{C}_n(X)$. At the corner where three points collide, the two adjacent boundary faces have opposite orientations. The cancellation is topological. No algebraic sign convention produces it.

Decomposing the differential by pole order clarifies the structure but destroys the computation. The Borchers identity governs all poles at once. The Jacobi identity governs one. When there are two poles, these are different identities.

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Equation (3) is the conformal anomaly. The classical BRST operator does not square to zero. The quantum correction restores it. The BV quantum master equation $\hbar\Delta S + \frac{1}{2}\{S, S\} = 0$ is the same equation.

The obstruction to $d^2 = 0$ is not an obstruction to the theory.

3 Three letters d

On $\mathbb{P}^1$ the propagator is single-valued and Arnold's relation holds. On an elliptic curve E_τ the function $z_1 - z_2$ is not single-valued.

Replace it by the odd theta function, with a correction forced by double-periodicity:

$$\eta_{12}^{(1)} = d \log \vartheta_1(z_1 - z_2 \mid \tau) + \frac{2\pi i \operatorname{Im}(z_1 - z_2)}{\operatorname{Im} \tau} d\bar{z}_1.$$

Arnold's relation fails. Of course it does—the propagator is no longer single-valued. What remains is the Arakelov $(1, 1)$ -form:

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g. \quad (4)$$

One number, $\kappa(\mathcal{A})$, extracted from the leading OPE coefficient, controls everything.

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Three differentials act on the genus- g bar complex.

(i) d_0 : genus-0 collision differential. Built from $d \log(z_i - z_j)$ on $\mathbb{P}^1$. Squares to zero by Arnold. This is classical Koszul duality.

(ii) d_{fib} : fiberwise differential on a fixed Σ_g . $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g \neq 0$. The world is curved.

(iii) D_g : total corrected differential, with A -period parameters $t_k = \oint_{A_k} \omega_k$. $D_g^2 = 0$.

The correction that restores nilpotence is a Lagrangian splitting. The A -cycles form a Lagrangian subspace of $H^1(\Sigma_g, \mathbb{C})$ for the intersection pairing; the curvature lies in the symplectic complement; the A -period corrections project it to zero. Nilpotence comes from the symplectic topology of the curve.

In the curved A_∞ framework: $m_0 = \kappa \cdot \omega_g$, and $m_1^2(a) = [m_0, a]$. The condition $\kappa = 0$ is simultaneously: $d_{\text{fib}}^2 = 0$; anomaly cancellation for the bosonic string; $c = 26$; the bar complex being a genuine derived object. When $\kappa \neq 0$ the bar complex is curved, and the curvature is the invariant.

At genus 1 the tadpole amplitude is $\kappa \cdot E_2(\tau)$. The Eisenstein series transforms as $E_2(-1/\tau) = \tau^2 E_2(\tau) + 6\tau/(\pi i)$, and the anomalous term is controlled by κ .

Across all genera, the generating function of the obstruction classes is

$$\sum_{g \geq 1} \kappa(\mathcal{A}) \cdot \lambda_g \cdot \hbar^{2g-2} = \kappa(\mathcal{A}) \cdot (\hat{A}(i\hbar) - 1).$$

The $\hat{A}$ -genus, from index theory. It has no business being here.

4 Fourier

The bar construction is a Fourier transform.

Classical Fourier	Chiral bar
Kernel: $e^{-2\pi i x \xi}$	Kernel: $\eta_{ij} = d \log(z_i - z_j)$
$\widehat{f * g} = \hat{f} \cdot \hat{g}$	$\mathbb{D}_{\text{Ran}} \circ \bar{B} \simeq \bar{B} \circ (-)^!$ (Thm A)
$\hat{f}(x) = f(-x)$	$\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Thm B)
$\ f\ ^2 = \ \hat{f}\ ^2$	$\mathcal{Q}_g(\mathcal{A}) \oplus \mathcal{Q}_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (Thm C)
Char. fn. $\rightarrow$ moments	$\kappa(\mathcal{A}) \rightarrow$ genus tower via $\hat{A}$ (Thm D)

When X is a point, the propagator vanishes, d_{res} reduces to the binary bracket, and you recover the classical bar construction of Loday–Vallette. Operadic Koszul duality is the zero-dimensional shadow.

For the Heisenberg on E_τ , the bar complex is the Fourier–Mukai kernel on $\text{Jac}(E_\tau) \times \text{Jac}(E_\tau)^\vee$. The chiral bar construction is the Fourier–Mukai transform, generalized from abelian to non-abelian.

The logarithm converts multiplication to addition. The bar construction converts the OPE (multiplicative, with poles) to the bar differential (nilpotent, additive). The cobar reverses this: it inserts delta-function currents at boundary strata. Verdier duality on $\overline{\mathcal{C}}_n(X)$ exchanges residues with inclusions, so bar and cobar are Verdier dual. Theorems A–D are four faces of this one duality.

At $n = 4$, the compactification $\overline{\mathcal{M}}_{0,5}$ is the del Pezzo surface $\text{Bl}_4(\mathbb{P}^2)$, with 10 boundary divisors. Five respect cyclic ordering. The relation $\partial^2 = 0$ on this surface yields the A_∞ pentagon identity. The coherence relations of homotopical algebra come from blowing up $\mathbb{P}^2$ four times. Why should they?

For Koszul algebras: $m_k = 0$ for $k \geq 3$. For interacting algebras: $m_3 \neq 0$, and the pentagon is the first obstruction to formality. Whether a chiral algebra is Koszul is a question about the geometry of a del Pezzo surface.

5 Nilpotence is the logarithm of periodicity

The classical logarithm converts multiplication to addition. The bar construction does the same thing: it converts the OPE (a product, with poles) to the bar differential (a sum, nilpotent). The last section presented this as analogy. It is not analogy.

Write $\log^{\text{cat}}(\mathcal{A}) := \bar{B}_X(\mathcal{A})$ and $\exp^{\text{cat}}(C) := \Omega_X(C)$. Then:

$\exp(\log(z)) = z$ on $ z - 1 < 1$	$\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ on the Koszul locus
$\log(1+z) = z - z^2/2 + \dots$	PBW spectral sequence: Taylor expansion of the log
Convergence radius	E_2 degeneration (Koszulness)
Analytic continuation past radius	Coderived category
$\log(ab) = \log(a) + \log(b)$	$Q_g(\mathcal{A}) + Q_g(\mathcal{A}^\dagger) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (Thm C)
$\log(1) = 0$	$\bar{B}(k)$ is acyclic
Single-valued on simply connected domain	$d_0^2 = 0$ at genus 0
Multi-valued on $\mathbb{C}^\times$; monodromy $2\pi i$	$d_{\text{fib}}^2 = \kappa \cdot \omega_g$ at genus $g \geq 1$
Residue at branch point	Curvature $\kappa(\mathcal{A})$
Leading coefficient of $\log(1+z) = z + \dots$	Modular characteristic $\kappa(\mathcal{A})$ (Thm D)

Every row is a theorem. Not by design.

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The PBW spectral sequence is the Taylor expansion. The filtration degree is the order of the expansion: the p -th filtered piece retains contributions from at most p points on X , the first p terms of the log series. The differential d_1 extracts the leading (binary OPE) term. Higher differentials d_r are higher-order corrections from r -fold collisions.

E_2 degeneration means the Taylor expansion terminates: the log is polynomial in the filtration variable. Off the Koszul locus, the series diverges. The coderived category is the analytic continuation—the enlargement of the derived category that makes the divergent log well-defined. Positselski’s acyclicity theorem is the statement that the analytic continuation exists.

The Koszul locus is the convergence radius. For KM: linear OPE poles, finite radius, convergent. For admissible-level affine algebras: the OPE singularity structure forces non-terminating higher differentials, the radius shrinks to zero, and only the completed category survives.

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The genus transition is the passage from single-valued to multi-valued.

On $\mathbb{P}^1$, $\log(z_1 - z_2)$ is single-valued. Arnold’s relation holds. $d_0^2 = 0$. No monodromy. This is Theorems A and B.

On Σ_g with $g \geq 1$, the function $z_1 - z_2$ is not single-valued. Its replacement $E(z_1, z_2)$, the prime form, has monodromy $\exp(2\pi i \oint_{B_k} \omega_j)$ around the B -cycles. The categorical logarithm inherits this monodromy. Arnold’s relation acquires a correction. $d_{\text{fib}}^2 = \kappa \cdot \omega_g \neq 0$. The curvature κ is the residue at the branch point.

The Gauss–Manin correction restores nilpotence: $D_g^2 = 0$. The total differential trivializes the multi-valued log on the universal cover of moduli space. The three differentials d_0, d_{fib}, D_g are the germ, the multi-valued lift, and the trivialization. Three levels of the same logarithm.

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Complementarity is the additivity of the logarithm.

$Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$. The left side is $\log(\mathcal{A}) + \log(\mathcal{A}^!)$. The right side is $\log(\mathcal{A} \otimes \mathcal{A}^!)$. Theorem C says the categorical logarithm is a homomorphism. The Lagrangian decomposition is the decomposition of the total log into two branches—one for each sheet of the multi-valued function.

The functional equation of the Riemann zeta relates $\zeta(s)$ and $\zeta(1-s)$ through the gamma factor. Theorem C relates $\log^{\text{cat}}(\mathcal{A})$ and $\log^{\text{cat}}(\mathcal{A}^!)$ through the center $\mathcal{Z}_{\mathcal{A}}$. These are the same pattern: a duality that decomposes total data into complementary halves.

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The characteristic hierarchy is the order expansion of the logarithm.

$\kappa(\mathcal{A})$: the leading coefficient. Determines the scalar package. Its exponential is the period $N = 24q/\text{gcd}(p, 24)$ for $c = p/q$. This is $\exp(2\pi i N c/24) = 1$ — the monodromy of the T-matrix is the exponential of the central charge.

$\Delta_{\mathcal{A}}$: the eigenvalue spectrum. Determines the periodicity profile $\Pi(\mathcal{A})$. Each T-matrix eigenvalue $\lambda_i = \exp(2\pi i(h_i - c/24))$ is the exponential of a conformal weight.

$\Theta_{\mathcal{A}}$: the complete logarithm. Its exponential is the modular functor: the action of Γ_g on conformal blocks.

One is the leading term of the next. κ is the trace of Δ , Δ is the eigenvalue decomposition of Θ . In complex analysis: the winding number, the set of periods, the full multi-valued function.

*

Quantization is exponentiation. Dequantization is taking the logarithm.

The passage from Lie algebra to Lie group is $\exp$: nilpotent structure (Jacobi identity) produces periodic structure ($\exp(2\pi i X) = 1$ for lattice elements). The bar construction reverses this: it extracts the nilpotent infinitesimal data from the periodic algebraic structure. The cobar exponentiates it back.

The screening charges Q_i of lattice vertex algebras have $Q_i^2 = 0$. This is the nilpotent datum. The quantum group relation $e_i^N = 0$ at $q = e^{2\pi i/N}$ is the exponential datum: the N -th power of the screening operator vanishes in the quantum group. The root of unity is the monodromy of the categorical logarithm around the N -th root.

The passage $Q^2 = 0 \Rightarrow e^N = 0$ is not metaphorical exponentiation. The braiding matrix R of $\mathcal{O}_q$ satisfies $R = \exp(r)$ where r is the classical r -matrix—which is the binary OPE residue in the bar differential. The quantum group is the Lie group of the chiral Lie algebra, and the bar construction is the map from group to algebra.

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The four theorems are the four properties of a logarithm:

- A: the logarithm exists;
- B: $\exp \circ \log = \text{id}$ on the convergence domain;
- C: the branch structure is controlled (complementary sheets);
- D: the leading coefficient determines the local behavior.

The five master conjectures are five aspects of one question: what is the categorical logarithm for factorization algebras on algebraic curves?

MC1: does it converge? MC3: what is the convergence domain? MC4: does it extend to infinite rank? MC5: do the physical and mathematical logarithms coincide? Periodicity: what is the monodromy group?

Not five problems. One object, five windows.

6 Two things called E_1

Which E_1 ?

On a smooth curve X :

$$E_\infty\text{-chiral} \subset P_\infty\text{-chiral} \subset E_1\text{-chiral}.$$

E_∞ -chiral = vertex algebra. Skew-symmetric bracket, unordered configuration spaces. Heisenberg, KM, Virasoro, $\mathcal{W}$ -algebras.

E_1 -chiral = drop skew-symmetry. R -twisted locality: $J_a(z)J_b(w) - R_{cd}^{ab}(z-w)J_d(w)J_c(z) = 0$. When $R = P$ (permutation): E_∞ . When $R \neq P$: ordered configuration spaces. Yangians, quantum lattice algebras.

Meanwhile, separately, in a different part of mathematics: the little n -disks operad E_n lives on real n -manifolds. A complex curve is real 2-dimensional. The chiral bar complex lives at $n = 2$ (Arnold relations). Chern–Simons on a 3-manifold lives at $n = 3$ (Tataro relations).

An E_1 -chiral algebra lives on a curve. A topological E_1 -algebra lives on an interval. These are different objects that share an overloaded name. On an interval there are no configuration spaces, no Arnold relation, no propagator. The subject does not become simpler. It ceases to exist.

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The Kazhdan–Lusztig equivalence sends $\widehat{\mathfrak{g}}_k$ -modules to $U_q(\mathfrak{g})$ -modules, with $q = e^{\pi i/(k+h^\vee)}$ and braiding by the universal R -matrix. Verdier duality on $\overline{C}_n(X)$ reverses the braiding: $R \mapsto R^{-1}$.

Without ordering: Verdier duality acts as $k \mapsto -k - 2h^\vee$ (the Feigin–Frenkel involution). With ordering: $R \mapsto R^{-1}$ (the Drinfeld–Kohno theorem).

So:

$$Y_h(\mathfrak{sl}_N)^{!,\text{ch}} \cong Y_{R^{-1},h}(\mathfrak{sl}_N)^{\text{ch}}.$$

This single identity unifies three theorems. Feigin–Frenkel is its E_∞ shadow. Drinfeld–Kohno is its monodromy. $\text{Ass}^! = \text{Ass}$ is its shadow at a point.

Lattice vertex algebras make the E_∞/E_1 transition visible. A 2-cocycle ε on a lattice Λ produces V_Λ^ε . When ε is symmetric: E_∞ . Otherwise: strictly E_1 . Koszul duality inverts the cocycle: $\varepsilon \mapsto \varepsilon^{-1}$. For even unimodular Λ : self-duality, $V_\Lambda^! \cong V_\Lambda$. The E_8 lattice vertex algebra is Koszul self-dual.

A Coisson algebra (Beilinson–Drinfeld) is a Poisson vertex algebra, which is a commutative $\mathcal{D}_X$ -algebra with λ -bracket. It is not a chiral algebra. The quantization tower runs:

$$\text{Coisson} \xrightarrow{\text{star product}} E_\infty\text{-chiral} \xrightarrow{R\text{-matrix}} E_1\text{-chiral}.$$

Holomorphic-topological gauge theories in 3d produce Coisson algebras. Their quantization lands at E_∞ , not E_1 . Reaching E_1 requires the KZ equations on a curve—a 2-dimensional phenomenon. Three dimensions is not enough.

From 3d: Chern–Simons on $\Sigma_g \times [0, 1]$ produces the WZW model on Σ_g . The perturbative expansion gives A_∞ structure on flat connections—the 3d version of the curved A_∞ on the genus- g bar complex. The curvature m_0 in 2d is the CS level in 3d.

From 4d: the Costello–Li holomorphic-topological twist of $\mathcal{N} = 2$ gauge theory on $\mathbb{C} \times \mathbb{C}^*$, reduced along $|w| = 1$, yields a factorization algebra on $\mathbb{C}$. When conformal symmetry survives: a chiral algebra. Otherwise: a factorization algebra that lives in the Costello–Gwilliam framework, outside the $\mathcal{D}$ -module world.

7 The BV complex is the bar complex

The BV bracket $\{-, -\}$ of degree +1 is the simple-pole OPE residue. The BV Laplacian Δ is the double-pole differential. The quantum master equation $\hbar\Delta e^{S/\hbar} = 0$ is bar-cobar Verdier compatibility. These are not analogies. They are identities.

The conformal anomaly is one object seen from three windows:

- the curvature element $m_0 = \kappa/2$ of the curved A_∞ ;
- the obstruction class in $R^*(\overline{\mathcal{M}}_g)$;
- the failure $\hbar\Delta S + \frac{1}{2}\{S, S\} \neq 0$ when $c \neq 26$.

For the bosonic string: $\kappa(\mathcal{H}^{26}) = 13$, $\kappa(bc) = -13$, total $\kappa = 0$. No curvature. No anomaly. The QME holds. These are the same sentence, said three ways.

The propagator transforms under coordinate changes as a BRST ghost for diffeomorphisms. The bar complex is a coordinate-free BRST complex. At genus 0 this is proved. At higher genus it requires Costello's renormalization machinery, and waits downstream.

8 Koszulness is a theorem, not a definition

$U(\mathfrak{g})$ is Koszul. This says nothing about $\widehat{\mathfrak{g}}_k$. The classical bar complex uses one product. The chiral bar complex uses the full OPE, all poles at once. Higher poles are not perturbations. The Sugawara tensor $T = \frac{1}{2(k+h^\vee)} \sum_a J^a J^a$: produces quartic poles in $T(z)T(w)$ via double Wick contractions. These have no classical analogue.

There is a circularity. To compute bar cohomology: assume Koszulness (spectral sequence collapses at E_2). Bar cohomology would then verify Koszulness (via $H_{\mathcal{A}}(t) \cdot H_{\mathcal{A}^!}(-t) = 1$). Breaking this circle requires an independent collapse proof, and every such proof is specific to its family.

KAC-MOODY. Whitehead's theorem ($H^n(\mathfrak{g}, V) = 0$ for $n = 1, 2$) handles the off-diagonal, and Killing contraction handles the invariant piece. The PBW enrichment at E_1 is $\mathfrak{g} \otimes \mathfrak{g}$ (dimension 9 for $\mathfrak{sl}_2$), not $\Lambda^2(\mathfrak{g})$ (dimension 3).

VIRASORO. The subalgebra $\{L_{-1}, L_0, L_1\} \cong \mathfrak{sl}_2$ acts independently of the central charge, because $m^3 - m = 0$ for $m \in \{-1, 0, 1\}$. The Casimir eigenvalues at every weight $h \geq 2$ are positive.

PRINCIPAL $\mathcal{W}_N$. Unique weight-2 mechanism from the stress tensor; triangular d_2 argument.

There is no universal collapse theorem. Each algebra must be convicted on its own evidence.

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Free fields (Heisenberg, free fermion) have one OPE pole: $d_{\text{bracket}} = 0$, no Borchers needed, no PBW needed. Their bar cohomology grows sub-exponentially—partitions, $e^{C\sqrt{n}}$. Interacting fields (KM, Virasoro, $\mathcal{W}$ -algebras) have two poles, and their bar cohomology grows exponentially— $3^n, 8^n$. The jump from one pole to two is visible in the asymptotics. One pole: garden. Two poles: wilderness.

9 What the bar complex counts

Algebra	Dims (1–6)	Growth	Sequence
Heisenberg	1, 1, 1, 2, 3, 5	$e^{C\sqrt{n}}$	partitions $p(n)$
Free fermion	1, 1, 2, 3, 5, 7	$e^{C\sqrt{n}}$	partitions $p(n-1)$
bc ghosts	2, 3, 6, 13, 28, 59	2^n	—
$\beta\gamma$	2, 4, 10, 26, 70, 192	$3^n/\sqrt{n}$	A025565
$\widehat{\mathfrak{sl}}_2$	3, 6, 15, 36, 91, 232	$3^n/n^{3/2}$	Riordan numbers
Virasoro	1, 2, 5, 12, 30, 76	$3^n/n^{3/2}$	Motzkin differences
$\widehat{\mathfrak{sl}}_3$	8, 36, 204, 1352	8^n	(conjectured)
$\mathcal{W}_3$	2, 5, 16, 52, 171	3.3^n	(conjectured)
$Y(\widehat{\mathfrak{sl}}_2)$	4, 10, 28, 82	3^n	$3^n + 1$ (conjectured)

The $\widehat{\mathfrak{sl}}_2$ generating function is

$$H(x) = \frac{1 + x - \sqrt{(1 - 3x)(1 + x)}}{2x(1 + x)}.$$

These are the Riordan numbers. Nobody asked for them. They count non-crossing partitions with a certain forbidden pattern. After Drinfeld–Sokolov reduction to the Virasoro algebra, the generating function rearranges into Motzkin differences $M(n+1) - M(n)$, lattice paths in the half-plane. Both come from the same discriminant $\Delta = (1 - 3x)(1 + x)$; the DS reduction is a change of variables that preserves the branch cut.

The discriminant is a DS-reduction invariant. The $\mathfrak{sl}_3$ family has discriminant $1 - 3x - x^2$, with roots involving $\sqrt{13}$ (and $13 = 3^2 + 4$). The recurrence depth of the generating function equals $\text{rank}(\mathfrak{g}) + 1$ in all known cases.

The conjectured Yangian formula $H^n(Y(\widehat{\mathfrak{sl}}_2)) = 3^n + 1$ would give the only interacting algebra with a rational generating function—no branch points at all. This may reflect the fact that Yangian OPE data is rational (Yang R -matrix), while KM data involves transcendental functions (theta, Weierstrass) at higher genus. Seven independent consistency checks support the conjecture through degree 10. If true, it demands an explanation.

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$\mathcal{W}$ -algebra Koszul duality for the principal nilpotent: $\mathcal{W}^k(\mathfrak{g})^\dagger \cong \mathcal{W}^{k'}(\mathfrak{g})$, where $k' = -k - 2h^\vee$. The proof goes through the Wakimoto free-field resolution: compute the bar complex of the free fields, observe that the BRST screening momenta reverse sign ($\alpha \rightarrow -\alpha$), forcing the dual level.

For $\mathfrak{sl}_2$: $k' = -k - 4$, and $c(k) + c(k') = 26$. For $\mathfrak{sl}_3$: $c + c' = 100$. The anomaly ratio $\rho(\mathfrak{g}) = \kappa/c = \sum (m_i + 1)^{-1}$, summed over exponents, is level-independent. For $\mathfrak{sl}_2$: $1/2$. For $\mathfrak{sl}_N$: the harmonic number $H_N - 1$. The genus-1 free energy is $\kappa/24$ for every principal $\mathcal{W}$ -algebra.

For non-principal nilpotent orbits, the expected duality is $\mathcal{W}^k(\mathfrak{g}, f)^\dagger \cong \mathcal{W}^{k'}(\mathfrak{g}, f^D)$, where f^D is the Barbasch–Vogan dual orbit. Chain-level evidence for all type- A hook pairs $(n - r, 1^r)$ with $n \leq 20$.

10 Five invariants, one number

The genus expansion of a Koszul chiral algebra $\mathcal{A}$ is controlled by a 5-tuple:

$$C_{\mathcal{A}} = (\Theta_{\mathcal{A}}, \kappa(\mathcal{A}), \Delta_{\mathcal{A}}, \Pi_{\mathcal{A}}, H_{\mathcal{A}}).$$

$\Theta_{\mathcal{A}}$: universal Maurer–Cartan class. Proved to exist. $\kappa(\mathcal{A})$: scalar modular characteristic. Proved; generating function is the $\hat{A}$ -genus. $\Delta_{\mathcal{A}}$: spectral discriminant. Proved; DS-invariant. $\Pi_{\mathcal{A}}$: periodicity package. Partly proved, partly open. $H_{\mathcal{A}}$: Hochschild duality, $\mathrm{HH}^n(\mathcal{A}) \cong \mathrm{HH}^{2-n}(\mathcal{A}^\vee) \otimes \omega_X$. Proved.

For any simple $\mathfrak{g}$, the class $\Theta_{\mathcal{A}}$ collapses to a single scalar. $\Theta_{\mathcal{A}}^{\min} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda$. One number, one kernel, one class.

Why: $H_{\mathrm{cyc}}^2(\mathfrak{g}, \mathfrak{g}) = H^3(\mathfrak{g}) = \mathbb{C}$ for every simple $\mathfrak{g}$. One-dimensional. The cubic Casimir of $\mathfrak{sl}_3$ lives in $H^5(\mathfrak{g})$ —degree 5 in cyclic cohomology. The Maurer–Cartan equation lives at degree 2. The degrees do not meet. Higher Casimirs cannot contribute, at any rank.

Genera 1 and 2 are rigid: no freedom beyond κ . At genus 3:

$$l_1(\theta_3 - \kappa \cdot \mu \otimes \lambda_3) = -\frac{1}{6} l_3(\theta_1, \theta_1, \theta_1),$$

the first correction, purely Killing. The l_2 term vanishes by Jacobi: the first nontrivial partition of 3 as a sum of smaller genera is $1 + 1 + 1$, and l_2 needs only two inputs.

The modular operator $\kappa(\lambda)$ on $\overline{\mathcal{M}}_g$ is nilpotent. $\kappa(\lambda)^{g-1} = 0$ on $\mathcal{M}_g$ (Graber–Vakil). At $g = 2$: zero already. At $g = 3$: nilpotency index ≤ 2 . The operator does not cycle. It dies.

II The frame atom

The Heisenberg algebra $\mathcal{H}_k$ sees all four theorems A–D and encounters none of the pitfalls. Not because it is simple, but because its OPE has one pole.

Bar cohomology at degree n counts partitions of $n - 2$: $1, 1, 1, 2, 3, 5, 7, 11, \dots$. The generating function is $\prod_{n \geq 2} (1 - q^n)^{-1}$, a piece of the reciprocal Dedekind eta. A modular form, already, at the simplest example.

$\mathcal{H}_k$ is not self-dual: $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$. The Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ on KM levels is not a mystery; it is Verdier duality. The $-k$ comes from dualizing the form; the $-2h^\vee$ from the canonical class of X . For a Koszul pair $(\widehat{\mathfrak{g}}_k, \widehat{\mathfrak{g}}_{-k-2h^\vee})$:

$$c + c' = 2 \dim(\mathfrak{g}),$$

independent of k . Central charge complementarity is Theorem C at the scalar level, and the Freudenthal–de Vries identity ensures the sum is always an integer.

At the critical level $k = -h^\vee$: the Sugawara tensor is undefined (not “ c diverges”), the bar complex is uncurved ($\kappa = 0$), and

$$H^0(\bar{B}(\widehat{\mathfrak{g}}_{-h^\vee})) \cong \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D)) :$$

functions on the space of Langlands-dualopers. The entire geometric Langlands programme, at one value of k .

H^1 gives 1-forms on opers; H^2 gives 2-forms. At H^3 the Cartan transgression kills the top Lie cohomology class via a nonzero d_4 differential.

Four subjects—representation theory, geometric Langlands, bar-cobar duality, classical limit of the BV complex—converge on one cochain complex with zero curvature.

At admissible levels $k = -h^\vee + p/q$, the bar complex acquires periodic CDG structure indexed by q -th roots of unity. This is the door to logarithmic CFT.

12 Category $\mathcal{O}$

The natural conjecture: $D^b(\mathcal{O}_{Y_h(\mathfrak{sl}_N)})$ is thickly generated by finite-dimensional evaluation modules.

This is false. Objects in thick⟨fd evals⟩ have finite-dimensional total cohomology. Yangian Verma modules do not. The obvious guess fails for the obvious reason.

Four approaches to the correct statement:

I (BGG). Do not resolve Vermas by fd modules. Resolve standards by objects *induced* from evaluation data. The Baxter TQ -relations should categorify: $0 \rightarrow M_{x+1} \rightarrow \mathbb{C}_2 \otimes M_x \rightarrow M_{x-1} \rightarrow 0$.

II (Heart capture). If every simple of a finite-length abelian A lies in a thick $T \subset D^b(A)$: then $T = D^b(A)$. Proved for $\mathfrak{sl}_2$ unconditionally; for $\mathcal{O}_{\text{poly}}$ at all N .

III (Truncation sectors). Restrict to finite truncation sectors of shifted Yangians. Each is highest-weight with finitely many irreducibles.

IV (Completion). Replace $D^b(\mathcal{O})$ by its completed enhancement $\hat{\mathcal{O}}$. Then $\text{Loc}(\langle \text{evaluations} \rangle) = \hat{\mathcal{O}}$ after completion. Infinite-dimensional standards arise as pro-limits of finite-length objects.

Strategies I and IV are not the same theorem.

For type A , the polynomial subcategory $\mathcal{O}_{\text{poly}}$ is thickly generated by evaluations (Drinfeld classification), and factorization DK is unconditionally proved:

$$\text{Fact}_{E_1}^{\text{fd}}(Y_h(\mathfrak{sl}_N)) \simeq \text{Fact}_{E_1}^{\text{fd}}(U_q(\mathfrak{sl}_N))^{\text{op}}.$$

The natural domain for general DK is the anti-dominantly shifted envelope $\mathcal{O}^{\text{sh} \leq 0}$, where ordinary $\mathcal{O}$ sits at shift $\mu = 0$ and negative prefundamentals sit at $\mu = -\alpha_i^\vee$.

13 The frontier

Five master conjectures. Approximately one hundred subsidiary claims hang from them.

MC1 (PBW). Proved: KM, Virasoro, principal $\mathcal{W}_N$. No universal mechanism.

MC2 (Characteristic package). Proved. Three sub-packages: chiral graph complex, modular-operadic MC, tautological-line support.

MC3 (DK/KL). DK-0/1 all types. DK-1 $\frac{1}{2}$ via lattice sectorwise finiteness. DK-2/3 type A and polynomial $\mathcal{O}$. Full $\mathcal{O}$ at $N \geq 3$ open; reformulated via completed localizing generation.

MC4 (Infinite towers). M-level complete for $\mathcal{W}_\infty$ and Yangian RTT. H-level comparison: six OPE coefficients in three local blocks.

MC5 (BV/BRST = bar). Genus 0 proved. Higher genus downstream of MC1–4.

The DK ladder:

DK-0	chain-level adjunction	proved
DK-1	evaluation-locus comparison	proved
DK-1 $\frac{1}{2}$	lattice sector	proved
DK-2/3	thick gen / factorization core	type A
DK-4/5	dg-shifted RTT / H-level	open

14 What the 1-form knows

The logarithmic 1-form $\eta_{ij} = d \log(z_i - z_j)$ is the unique meromorphic 1-form with conformal invariance, canonical residues, and the Arnold relation.

The Borchers identity governs all OPE poles at once. The Jacobi identity governs one.

The PBW enrichment is $\mathfrak{g} \otimes \mathfrak{g}$, not $\Lambda^2(\mathfrak{g})$.

The $\hat{A}$ -genus and the Feigin–Frenkel involution are not in the OPE data. They live in the geometry of $\overline{\mathcal{C}}_n(X)$.

E_1 -chiral algebras live on curves. Topological E_1 -algebras live on intervals. On an interval there is no propagator.

$H_{\text{cyc}}^2(\mathfrak{g}, \mathfrak{g}) = \mathbb{C}$ for every simple $\mathfrak{g}$. One dimension. The cubic Casimir cannot contribute to Maurer–Cartan at any rank.

$\kappa(\lambda)^{g^{-1}} = 0$ on $\mathcal{M}_g$.

Bar cohomology of $\widehat{\mathfrak{sl}}_2$ is Riordan numbers. Virasoro is Motzkin differences. The discriminant is a DS-reduction invariant.

The A_∞ pentagon lives on $\text{Bl}_4(\mathbb{P}^2)$.

The BV bracket is the simple-pole OPE residue. The BV Laplacian is the double-pole differential. Ghosts are propagators.

$\text{thick}\langle \text{fd evals} \rangle \neq D^b(\mathcal{O})$, but $\mathcal{O}_{\text{poly}}$ works, and the shifted envelope $\mathcal{O}^{\text{sh} \leq 0}$ is the natural completed domain.

At $k = -h^\vee$: $\kappa = 0$, no curvature, and $H^0(\bar{B}(\widehat{\mathfrak{g}}_{-h^\vee})) = \text{Fun}(\text{Op}_{\mathfrak{g}^\vee}(D))$.

The genus expansion of every Koszul chiral algebra is $\kappa(\mathcal{A}) \cdot (\hat{A}(i\hbar) - 1)$.

* * *

A 1-form with a simple pole.

A relation among three.

The rest follows.

15 NF-I: the H^4 barrier

The nuclear frontier problem NF-I asks for an independent verification of $H^4(\bar{B}(\widehat{\mathfrak{sl}}_3)) = 1352$. Every current piece of evidence is *indirect*: the rational generating function

$$H_{\bar{B}}(x) = \frac{1-x}{(1-8x)(1-3x-x^2)}$$

determines the sequence $[1, 8, 36, 204, 1352, 9892, 76084, 598592, \dots]$ from H^0 through H^3 plus the *ansatz* that the denominator is quadratic in x . Everything below records what we tried, what we proved, and where we got stuck.

15.1 What is proved

- (i) **Koszul dual from series inversion.** The identity $H_{\bar{B}}(x) \cdot H_{A^\dagger}(-x) = 1$ determines the Koszul dual Hilbert series uniquely from the bar Hilbert series. Writing $b_n = \dim(A^\dagger)_n$:

$$b_n = - \sum_{k=1}^n (-1)^k \dim H^k(\bar{B}) \cdot b_{n-k}, \quad b_0 = 1.$$

With $H^n = [1, 8, 36, 204, 1352, 9892, 76084, 598592]$ this gives

$$b_n = [1, 8, 28, 140, 392, 2884, 2716, 75488].$$

All positive through degree 7. The *chiral excess* $\varepsilon_n := b_n - \binom{8}{n}$ is $[0, 0, 0, 84, 322, \dots]$: Koszul duality agrees with $\Lambda(\mathfrak{g}^*)$ (exterior algebra) through degree 2 and diverges at degree 3.

- (ii) **Quadratic relation scan.** For a quadratic algebra $A = T(V)/(R)$ with $R \subseteq V \otimes V$, the Koszul dual is $A^\perp = T(V^*)/(R^\perp)$ and $(A^\perp)_n = V^{\otimes n}/I_n$ where I_n is the degree- n ideal generated by $R^\perp$. Scanning four relation spaces for $V = \mathfrak{g}$ ($\dim = 8$):

R	$\dim R$	$(A^\perp)_2$	$(A^\perp)_3$	$(A^\perp)_4$
$\Lambda^2(V)$	28	28	56	70
$\Lambda^2(V) + \langle \kappa \rangle$	29	29	...	...
$\Lambda^2(V) \oplus \text{diag}$	36	36	120	...
$V \otimes V$	64	64	512	4096

Only $R = \Lambda^2(V) \oplus \text{diag}$ reproduces $H^2 = 36$, but gives $H^3 = 120 \neq 204$. The *gap* $204 - 120 = 84$ equals the chiral excess $\varepsilon_3 = 140 - 56 = 84$. This is not a coincidence: both measure the same beyond-quadratic OPE contribution.

Conclusion: No choice of quadratic $R \subseteq V \otimes V$ reproduces the chiral bar cohomology beyond degree 2. The chiral OPE is not a quadratic algebra in the classical sense. The higher-pole data (double pole / Killing contraction) contributes non-quadratically to the bar differential.

- (iii) $d_{\text{bracket}}^2 \neq 0$. The bracket-only differential d_{bracket} on $\mathfrak{g}^{\otimes n} \otimes \text{OS}^{n-1}$ satisfies $d_{\text{bracket}}^2 \neq 0$. This was verified computationally for *all* 2048 sign conventions on $\mathfrak{g}^{\otimes n} \otimes \text{OS}^{n-1}$ (degrees 3–5, $\mathfrak{sl}_2$). The rank of d_{bracket}^2 is 8 (surjective onto B^1). The full differential $d = d_{\text{bracket}} + d_{\text{curvature}}$ has $d^2 = 0$ via the Borchers identity, *not* the Jacobi identity alone.

- (iv) **Inverse vacuum character.** The coefficients $c_H = [q^H] \prod_{n \geq 1} (1 - q^n)^8$ are

$$c_H = [1, -8, 20, 0, -70, 64, 56, 0, -125, \dots].$$

The convolution $\sum_{j=0}^H c_j \cdot V_{H-j} = \delta_{H,0}$ is verified through weight 8 (with V_H the vacuum character coefficients).

- (v) **Bracket ranks on tensor algebra.** The CE-type bracket $d: \mathfrak{g}^{\otimes n} \rightarrow \mathfrak{g}^{\otimes(n-1)}$ has ranks $d_2 = 8$, $d_3 = 56$, $d_4 = 456$, $d_5 = 3640$, and the tensor algebra is *exact* for semisimple $\mathfrak{g}$: $\text{rank}(d_{n+1}) = \ker(d_n)$ for $n \geq 2$. This is the CE complex on $T^*(\mathfrak{g})$, not on $\Lambda^*(\mathfrak{g})$, and is automatically acyclic.

- (vi) **Casimir decomposition.** $3C_2$ eigenspaces on $\mathfrak{g}^{\otimes n}$ for $\mathfrak{g} = \mathfrak{sl}_3$:

$$\begin{aligned} n = 2 \text{ (dim 64)} : & \{0 : 1, 18 : 16, 36 : 20, 48 : 27\}, \\ n = 3 \text{ (dim 512)} : & \{0 : 2, 18 : 64, 36 : 80, 48 : 162, 72 : 140, 90 : 64\}, \\ n = 4 \text{ (dim 4096)} : & \{0 : 8, 18 : 256, 36 : 400, 48 : 891, 72 : 1050, 90 : 768, \\ & 108 : 112, 120 : 486, 144 : 125\}. \end{aligned}$$

Bracket rank by eigenspace at $n = 4$: $\{0 : 1, 18 : 56, 36 : 60, 48 : 135, 72 : 140, 90 : 64, 108 : 0, 120 : 0, 144 : 0\}$. The three highest eigenspaces (108, 120, 144) have bracket rank 0 — they are entirely in the kernel. Invariant tensor dims: 0, 1, 2, 8 at $n = 1, 2, 3, 4$.

15.2 Where we are stuck: the $R^{(1)}$ barrier

The derivative-residue operator $R_{ij}^{(1)}$ acts on OS forms by:

$$R_{12}^{(1)}(\eta_{12} \wedge \eta_{13}) = -\frac{dw}{w^2}$$

which lies *outside* the Orlik–Solomon algebra. No finite-dimensional extension of OS by “double-pole forms” closes under $R^{(1)}$, because $R^{(1)}$ generates forms with arbitrarily high pole order under iteration.

This barrier manifests at **two levels**:

Differential level. One cannot build the full chiral bar differential $d = d_{\text{bracket}} + d_{\text{curvature}}$ as a matrix on $\mathfrak{g}^{\otimes n} \otimes \text{OS}^{n-1}$ because $d_{\text{curvature}}$ maps outside this space. The bracket-only truncation has $d_{\text{bracket}}^2 \neq 0$, so it cannot be used alone.

Chain-level (Euler characteristic obstruction). Neither naive model of the bar complex is consistent with $H^n = [1, 8, 36, 204]$:

- *Model A* ($\mathfrak{g}^{\otimes n} \otimes \text{OS}^{n-1}$): The weight-2 chain Euler characteristic is 56. Since each H^n component at weight 2 contributes non-negatively, this forces $H_2^2 \geq 56 > 36 = H^2$. Contradiction.
- *Model B* ($\bar{A}^{\otimes n} \otimes \text{OS}^{n-1}$, with $\bar{A} = \bigoplus_{h \geq 1} V_h$): The weight-3 chain Euler characteristic is -512 . By alternating-sign accounting this forces $H_3^3 \geq 512 > 204 = H^3$. Contradiction.

Both models fail because the true chain complex involves the $R^{(1)}$ operator mixing OS degrees with pole orders, and neither truncation captures this correctly.

15.3 The PBW paradox and its resolution

The PBW spectral sequence has $E_1^{p,q}$ with $E_1^{1,0} = \bar{V}^{\mathfrak{g}} = 0$ at every weight (since $\mathfrak{sl}_3$ is semisimple, invariants in $\bar{V}_h$ vanish for $h \geq 1$). Yet $H^1(\bar{B}(\widehat{\mathfrak{sl}}_3)) = 8 \neq 0$.

Resolution: the PBW spectral sequence converges to the Koszul complex $K = \bar{B}(\mathcal{A}) \otimes_{\tau} \mathcal{A}$ (which is acyclic), *not* to the bar complex $\bar{B}(\mathcal{A})$ itself. The E_{∞} page records acyclicity of K , and the bar cohomology is extracted differently — via the Hilbert series identity $H_{\bar{B}}(x) \cdot H_{\mathcal{A}}(-x) = 1$ on the Koszul locus.

This is why the PBW approach to NF-I is indirect: it proves *acyclicity of the Koszul complex*, from which one extracts bar cohomology dimensions via series inversion against the (independently known) vacuum character. It does not give H^4 without already knowing H^0 through H^3 .

15.4 What remains for NF-I

The following approaches have been **ruled out**:

- Direct matrix computation* of d on $\mathfrak{g}^{\otimes n} \otimes \text{OS}^{n-1}$: blocked by $d_{\text{bracket}}^2 \neq 0$ and the $R^{(1)}$ barrier.
- Quadratic algebra models*: no $R \subseteq V \otimes V$ reproduces chiral bar cohomology beyond degree 2.
- PBW direct readoff*: the spectral sequence converges to the Koszul complex, not the bar complex.

Viable approaches (in decreasing order of concreteness):

- (i) **Euler characteristic bootstrap.** If the weight decomposition $H^n = \bigoplus_h H_h^n$ were known for $n \leq 3$, the alternating Euler characteristic at each weight constrains H_h^4 and may determine it. The weight- h vacuum character provides $\chi_h = \sum (-1)^n \dim H_h^n$. Currently we know χ_h (from the inverse vacuum character) but not the individual H_h^n . A “weight spectral sequence” decomposing each H^n by conformal weight would unlock this.
- (ii) **Zhu algebra / BRST functor.** Zhu’s algebra $A(\mathcal{A})$ captures H^0 ; a higher Zhu functor for H^n might give H^4 from representation-theoretic data without building the full bar complex. This is essentially MC_5 (BV = bar) restricted to degree 4.
- (iii) **Factorization homology.** On $\mathbb{P}^1$ with n marked points, the factorization structure gives H^n as the cohomology of a configuration-space integral. For $n = 4$ this is an integral over $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, which is low-dimensional enough to be computable. The obstacle is making the integrand (involving all OPE poles) explicit.
- (iv) **Denominator bootstrap.** The rational generating function ansatz $H_{\bar{B}}(x) = P(x)/Q(x)$ with $\deg Q = 2$ determines H^4 from $H^0, \dots, H^3$ uniquely. *Proving* that $\deg Q \leq 2$ would verify $H^4 = 1352$. Evidence: the vacuum character has a product formula $\prod (1 - q^n)^{-8}$, and the bar Hilbert series inverts it, so $H_{\bar{B}}$ is a rational function of the Dedekind eta function. The denominator degree may follow from the structure of the PBW filtration (two-step: bracket + Killing).

15.5 The 84

The number 84 appears in three independent ways:

$$\begin{array}{ll}
 204 - 120 = 84 & (\text{chiral } H^3 \text{ minus quadratic-model } H^3), \\
 140 - 56 = 84 & (\text{Koszul dual } (A^1)_3 \text{ minus } \binom{8}{3}), \\
 84 = \dim(\text{ad}^{\otimes 3})_{\text{beyond-antisym}}^8 & (\text{invariant tensors beyond } \Lambda^3).
 \end{array}$$

This is the “chiral excess”: the contribution of the double pole (Killing form / level) to degree-3 bar cohomology, visible both in the bar complex and in the Koszul dual. The excess $\varepsilon_n = b_n - \binom{8}{n}$ measures how far the chiral algebra departs from a classical universal enveloping algebra. At degree 3 it equals the dimension of symmetric cubic invariants of $\mathfrak{sl}_3$ (the “ d -symbol” space), which is 84-dimensional in the tensor algebra model.

At degree 4 the excess is $\varepsilon_4 = 392 - 70 = 322$. An independent computation of this number from OPE data would effectively verify $H^4 = 1352$.

* * *

Remark 15.1 (The shape of the barrier). The $R^{(1)}$ barrier is not a failure of technique but a *structural feature* of chiral algebras: the OPE is not a binary operation on a fixed vector space, but a family of operations parametrized by the formal disc, with poles of all orders contributing. Any finite-dimensional truncation loses information. The bar complex is infinite-dimensional at each cohomological degree (as a graded vector space over conformal weights), and the differential mixes all weights via the curvature term. This is why the Borcherds identity — which packages all poles simultaneously — is essential, and why purely combinatorial (matrix-rank) approaches fail.

The correct framework is the one in the monograph: configuration space integrals on FM compactifications, where the poles become boundary strata and $d^2 = 0$ follows from the geometry of the compactification, not from algebraic identities on a finite-dimensional chain complex.

16 Frontier assessment, March 2026

16.1 What is proved (Stratum I)

- (i) **Theorems A/B/C/D.** Bar-cobar adjunction (A), inversion on Koszul locus (B), complementarity (C), and both modular characteristics (D_{scal} and D_{Δ}). All four with complete proofs.
- (ii) **MC1 (PBW concentration).** Proved for KM (all simple $\mathfrak{g}$), Virasoro, and principal W_N . Computational verification: $\mathfrak{sl}_2$ through weight 8, $\mathfrak{sl}_3$ Casimir decomposition through $n = 4$.
- (iii) **MC2 (cyclic $L_{\infty} + \Theta_A$).** All three packages resolved. Both `conj:universal-theta` and `conj:universal-MC` upgraded from conjectured to proved. Scalar saturation proved for all simple $\mathfrak{g}$ (since $H_{\text{cyc}}^2(\mathfrak{g}, \mathfrak{g}) = H^3(\mathfrak{g}) = \mathbb{C}$ is 1-dimensional).
- (iv) **DK ladder: rungs 0–1½–3.** DK-0 (vacuum), DK-1 (chain-level), DK-1½ (lattice sector), DK-2/3 (finite-dimensional type A + eval core at all N). Category $\mathcal{O}_{\text{poly}}$ works at all ranks.
- (v) **Theorem H (Hochschild duality).** $\text{HH}^n(\mathcal{A}) \cong \text{HH}^{2-n}(\mathcal{A}^!)^{\vee} \otimes \omega_X$.
- (vi) **E1 sub-exponential growth (all simple types).** Corrected the polynomial growth claim: $\dim E_1^{0,p}(\mathfrak{g}[t]) \sim C(\mathfrak{g}) \cdot p^{-3/4} \cdot \exp(\pi\sqrt{rp}/12)$ where $r = \text{rank}(\mathfrak{g})$. Verified computationally for all 17 simple types through $p = 500$. Growth constant $\pi\sqrt{r}/12$ depends only on rank.
- (vii) **W_{∞} stage-4 structural closure.** Full reduction chain $29 \rightarrow 7 \rightarrow 6 = 4 + 2$ verified: primaryity, top-pole extraction, parity compression, OPE block decomposition. Four free coefficients, two residue checks. 72 tests, all passing.
- (viii) **Yangian M-level identification.** $K^{\text{line}} = K^{\text{RTT}}$ at M-level. RTT truncation defect: $\delta_n = \dim(Y_n) - \dim(Y_n^{\text{RTT}}) = [0, 0, 0, 3, 20, 87]$, matching the Serre ideal growth.
- (ix) **E1 inversion principle.** Lattice $\mathcal{E} \rightarrow \mathcal{E}^{-1}$ and Yangian $R \rightarrow R^{-1}$ are instances of the same bar-cobar inversion.
- (x) **Geometric depth.** $\kappa(\lambda)^{g-1} = 0$ on M_g (sharp, from Graber–Vakil). Nilpotent, not periodic.

16.2 Where we are stuck

Five specific bottlenecks, ordered by proximity to resolution.

1. MC4 H-level coefficient extraction. The M-level identification $K^{\text{line}} = K^{\text{RTT}}$ is proved, but the H-level (derived/infinity-categorical) comparison requires extracting specific coefficients from the homotopy transfer. For Yangians: the RTT truncation defect $\delta_n = [0, 0, 0, 3, 20, 87]$ gives the Serre ideal corrections but their *signs* in the A_{∞} structure maps m_n are not yet determined. For W_{∞} : the stage-4 frontier has 4 free coefficients ($c_{334}, c_{444}, C_{3,4;3;0,4}, C_{3,4;4;0,3}$) that require explicit OPE computation at $\text{spin} \geq 3$.

2. Category $\mathcal{O}$ thick generation beyond Rep_{fd} . $D^{\text{eval}} = D^b(\text{Rep}_{\text{fd}})$ exactly — this is *smaller* than $D^b(\mathcal{O})$. Extending DK beyond Rep_{fd} requires either: (a) $\mathcal{O}_{\text{poly}}$ thick generation (proved sectorwise, globalisation open), or (b) Baxter Q -operators for non-polynomial modules. The “completed” or “coderived” enhancement $D^{\text{co}}(\mathcal{O})$ may be the correct target, but its construction is open.

3. Completed/coderived formalism (DK-4/5). The coderived category $D^{\text{co}}(\mathcal{A}\text{-mod})$ should replace D^b when $\mathcal{A}$ has infinite-dimensional weight spaces. Positselski’s framework applies but the chiral adaptation (completed tensor products on Ran , convergent bar complexes) is not written.

4. MC5 higher genus. BV/BRST = bar at genus 0 is proved. At genus $g \geq 1$, the identification requires: (a) the BV bracket on higher-genus factorization homology, (b) compatibility of bar curvature $\kappa \cdot \omega_g$ with BV anomaly. Both are downstream of MC2-4.

5. Periodicity programme. Nilpotent (not periodic): $\kappa(\lambda)^{g-1} = 0$ on M_g . “Periodicity” was a misnomer. The correct question is: what is the nilpotency index of $\sigma_{\mathcal{A}}$ on $\bar{M}_g$, and how does it relate to $\text{lcm}(N_{\text{mod}}, N_{\text{quant}})$? Orthogonal to the main MC chain; not a bottleneck but conceptually open.

16.3 Concrete insights from computation

These are hard-won facts, each verified by independent computation and cross-checked against the manuscript.

- (a) **Polynomial growth was wrong.** The original claim in `conj:mc3-sectorwise-all-types` that $\dim E_1^{0,p} = O(p^N)$ is *false*. At $p = 200$ for $\mathfrak{sl}_2$: $\dim = 159,733$, apparent polynomial degree 2.26 (and increasing). The correct asymptotics are $\exp(\pi\sqrt{p/12})$, sub-exponential but not polynomial. This changes the Strategy IV analysis: sectorwise finiteness of E_∞ must come from *cancellation in differentials*, not from E_1 boundedness.
- (b) **The W_4 reduction chain is exact.** Starting from $|I_4| = 54$ primary seed OPE coefficients: primaryity reduces to $|J_4| = 29$ (Virasoro elimination); top-pole extraction to $|J_4^{\text{top}}| = 7$; parity compression to $|J_4^{\text{par}}| = 6$; OPE block decomposition: $6 = 1 + 3 + 2$ (blocks $(3, 3), (3, 4), (4, 4)$). Mixed/self split: 4 free + 2 checks. The two checks are: $C_{4,4;2;0,6}^{\text{res}} = 2$ (from $c = 3 - 60(k+3)^2/(k+4)$) and $C_{3,4;2;0,5}^{\text{res}} = 0$ (Virasoro decoupling).
- (c) **Complementarity sum 246.** For W_4 : $c(k) + c(k') = 246$ where $k' = -k - 8$. The number $246 = \dim(E_8) - 2$ is not a coincidence — it reflects the Feigin–Frenkel duality for the DS reduction of $\mathfrak{sl}_4$.
- (d) **The $d_{\text{bracket}}^2 \neq 0$ barrier is absolute.** Not a sign error: all 2048 conventions fail. Not a model error: $R^{(1)}$ maps outside OS. The *correct* framework is the geometric one (FM compactifications, Verdier duality), where $d^2 = 0$ follows from the compactification geometry.
- (e) **E_1 departure point identifies the second exponent.** For $\mathfrak{g}$ with exponents $e_1 = 1 < e_2 < \dots$, the first departure of $E_1(\mathfrak{g}[t])$ from $E_1(\mathfrak{sl}_2[t])$ occurs at $p = 2e_2 + 1$. For $\mathfrak{sl}_3$: $p = 5$ (second exponent $e_2 = 2$). For G_2 : $p = 11$ (second exponent $e_2 = 5$). This is a computable invariant that detects rank and exponents from the spectral sequence alone.
- (f) **Growth constant is rank-only.** $\pi\sqrt{r/12}$ for all simple $\mathfrak{g}$ of rank r , regardless of exponents. Same-rank algebras (e.g. A_2, B_2, C_2, G_2) have identical leading growth but different subleading terms. The subleading structure detects the full exponent sequence.

16.4 Next moves

In order of expected impact:

1. **MC4 Yangian: determine A_∞ signs.** The truncation defect δ_n gives sizes; the homotopy transfer theorem gives a formula for signs. This is a finite computation for each n . Target: m_3 signs through degree 6.
2. **MC4 W_∞ : compute the four free coefficients.** Explicit spin-3/spin-4 OPE at generic central charge. Each coefficient is a rational function of c . The two residue checks ($C_{4,4;2}^{\text{res}} = 2, C_{3,4;2}^{\text{res}} = 0$) provide falsifiable predictions.

3. **MC₃ Strategy IV: E_∞ convergence.** Now that E_1 growth is proved sub-exponential (not polynomial), the key question is whether differentials d_r produce enough cancellation for E_∞ sectorwise finiteness. A computational attack: compute d_2 for $\mathfrak{sl}_2[t]$ at small p and measure the kernel growth.
4. **DK-4: coderived enhancement.** Write the chiral adaptation of Positselski’s coderived category. The main technical point: completed tensor products on Ran preserve the bar-cobar adjunction.
5. **NF-1: denominator bootstrap.** Proving $\deg Q \leq 2$ for $H_{\bar{B}}(x) = P(x)/Q(x)$ would give $H^4(\bar{B}(\widehat{\mathfrak{sl}}_3)) = 1352$ and all higher degrees. The evidence is strong: PBW two-step filtration (bracket + Killing) suggests $Q(x)$ has degree ≤ 2 .

16.5 The shape of the problem

Every open question reduces to one phenomenon: the categorical logarithm has branch cuts. At genus 0, the logarithm is single-valued (Theorems A/B). At genus $g \geq 1$, the curvature $\kappa \cdot \omega_g$ introduces monodromy, the spectral sequence has non-trivial differentials, and the growth of E_1 reflects the density of branch points.

The sub-exponential growth $\exp(\pi\sqrt{rp/12})$ is the partition function of these branch points — it counts the ways to distribute curvature insertions across the bar complex. The E_∞ convergence question (MC₃) asks whether the monodromy is “tame” enough for the logarithm to converge.

The $d_{\text{bracket}}^2 \neq 0$ barrier says: you cannot take the logarithm term-by-term. The Borchers identity says: you can take it all at once, geometrically. This is the content of Theorem A.

What remains is to understand the domain of convergence (MC₃), the infinite-rank limit (MC₄), and the physical identification (MC₅). All three are aspects of one question: what is the categorical logarithm, completely?